Independent solutions

Expectation, laws, tail integrals, and finite conditioning

Use this document only after a complete attempt. Compare the first step at which your argument ceases to be legitimate; do not replace the work record with a cleaner proof.

Exercise 0 – The L^p–RN retention gate

Since $\mathbb{P}(\Omega) = 1$ and $q > 1$, the finite-measure embedding gives

$$\|Y\|_1 \le \|Y\|_q,$$

so $Y \in L^1(\mathbb{P})$. The numerator is the restriction of the finite signed measure $Y\mathbb{P}$ to $(\Omega, \mathcal{G})$; the reference is $\mathbb{P}|_{\mathcal{G}}$; both live on $(\Omega, \mathcal{G})$. Moreover,

$$|\nu_Y^{\mathcal{G}}|(\Omega) \le \mathbb{E}[|Y|] < +\infty,$$

and $\mathbb{P}(A) = 0$ implies $\nu_Y^{\mathcal{G}}(A) = 0$.

The signed Radon–Nikodym theorem therefore supplies a $\mathcal{G}$-measurable

$$H \in L^1(\mathbb{P}|_{\mathcal{G}})$$

such that

$$\nu_Y^{\mathcal{G}}(A) = \int_A H \, d\mathbb{P}, \qquad A \in \mathcal{G},$$

with H unique $\mathbb{P}$-almost surely. This representative is $\mathbb{E}[Y \mid \mathcal{G}]$.

For a random element X, LOTUS is the pushforward integration identity for the measure $\mu_X = X_\#\mathbb{P}$. It therefore needs no dominating Lebesgue measure. A formula involving f_X additionally requires $\mu_X \ll \lambda$; only then does Radon–Nikodym provide $f_X = d\mu_X/d\lambda$, unique λ-almost everywhere.

Exercise 1 – Expectation through the law

For $C \in \mathcal{S}$,

$$\mathbb{E}[\mathbf{1}_C(X)] = \mathbb{P}(X \in C) = \mu_X(C) = \int_S \mathbf{1}_C \, d\mu_X.$$

If $\varphi = \sum_{k=1}^n a_k \mathbf{1}_{C_k}$ is nonnegative simple, linearity gives

$$\mathbb{E}[\varphi(X)] = \sum_{k=1}^n a_k \mu_X(C_k) = \int_S \varphi \, d\mu_X.$$

For nonnegative measurable φ, choose $\varphi_n \uparrow \varphi$ with φ_n simple. Monotone convergence on S and on Ω yields

$$\mathbb{E}[\varphi(X)] = \lim_n \mathbb{E}[\varphi_n(X)] = \lim_n \int_S \varphi_n \, d\mu_X = \int_S \varphi \, d\mu_X.$$

For signed φ, apply this result to φ^+ and φ^- whenever their difference is defined.

On $((0,1), \mathcal{B}((0,1)), \lambda)$, the nonnegative variable $X(t) = 1/t$ has infinite expectation. For an undefined signed expectation, take the identity variable on $\mathbb{R}$ under the standard Cauchy probability law: both $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ are infinite.

If μ_X is atomic, integration against it is a weighted sum. If $\mu_X \ll \lambda$ with density f_X, the same integral is represented by $\int \varphi(x) f_X(x) \, dx$. Neither representation changes the definition of expectation.

Exercise 2 – Subgraphs, tail integration, and insurance payments

For $q \in \mathbb{Q}_{>0}$, both $\{f > q\} \in \mathcal{A}$ and $(0, q) \in \mathcal{B}((0, +\infty))$. The identity

$$H_f = \bigcup_{q \in \mathbb{Q}_{>0}} \{f > q\} \times (0, q)$$

therefore proves product measurability. Indeed, if $t < f(x)$, density of the rationals supplies q with $t < q < f(x)$; the converse is immediate.

Because $\mathbf{1}_{H_f} \ge 0$, Tonelli applies without any prior integrability claim:

$$\begin{aligned}
\int_E f(x) \, \mu(dx) &= \int_E \int_0^\infty \mathbf{1}_{\{t < f(x)\}} \, dt \, \mu(dx) \\
&= \int_0^\infty \int_E \mathbf{1}_{\{f(x) > t\}} \, \mu(dx) \, dt \\
&= \int_0^\infty \mu(f > t) \, dt.
\end{aligned}$$

Apply the same identity to the transformed variables. Since

$$\mathbb{P}((X-d)_+ > t) = \mathbb{P}(X > d+t),$$

the change of variable $s = d+t$ gives

$$\mathbb{E}[(X-d)_+] = \int_d^\infty \mathbb{P}(X > s)\, ds.$$

Since $\mathbb{P}(X \wedge u > t) = \mathbb{P}(X > t)$ for $0 \le t < u$ and is zero for $t \ge u$,

$$\mathbb{E}[X \wedge u] = \int_0^u \mathbb{P}(X > t)\, dt.$$

If $X \sim \mathrm{Exp}(\lambda)$, these are

$$\frac{e^{-\lambda d}}{\lambda} \quad \text{and} \quad \frac{1-e^{-\lambda u}}{\lambda}.$$

For the limited-loss payment,

$$\mathbb{E}[\min\{(X-d)_+, u\}] = \int_d^{d+u} \mathbb{P}(X > s)\, ds.$$

Under the exponential law this equals

$$\frac{e^{-\lambda d} - e^{-\lambda(d+u)}}{\lambda} = \frac{e^{-\lambda d}(1-e^{-\lambda u})}{\lambda}.$$

If g is convex and both X and $g(X)$ are integrable, Jensen gives

$$g(\mathbb{E}[X]) \le \mathbb{E}[g(X)].$$

It supplies a bound. It does not evaluate the tail integral defining the expected payment.

Exercise 3 – Conditioning on a finite partition

Let $\mathcal{G} = \sigma(B_1, \dots, B_r)$. A finite real-valued $\mathcal{G}$-measurable random variable is constant on each atom, so it has the form

$$H = \sum_{i=1}^r c_i \mathbf{1}_{B_i}.$$

Testing the conditional-expectation identity on B_j gives

$$c_j \mathbb{P}(B_j) = \mathbb{E}[Y \mathbf{1}_{B_j}].$$

When $\mathbb{P}(B_j) > 0$, this determines

$$c_j = \frac{\mathbb{E}[Y \mathbf{1}_{B_j}]}{\mathbb{P}(B_j)}.$$

Hence

$$\mathbb{E}[Y \mid \mathcal{G}] = \sum_{i=1}^r \frac{\mathbb{E}[Y \mathbf{1}_{B_i}]}{\mathbb{P}(B_i)} \mathbf{1}_{B_i}.$$

Taking expectations yields

$$\mathbb{E}[Y] = \sum_{i=1}^r \mathbb{P}(B_i) \mathbb{E}[Y \mid B_i].$$

On a null atom, the equation is $0 = 0$, so its coefficient is arbitrary. This is precisely version freedom on a null set. Taking $Y = \mathbf{1}_A$ gives

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i),$$

the law of total probability.

Exercise 4 – Bayes and representation bookkeeping

The joint masses satisfy

$$\mathbb{P}(A \cap B_i) = \mathbb{P}(A \mid B_i) \mathbb{P}(B_i).$$

Summing over the partition gives

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i).$$

Therefore

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

For $C \in \mathcal{F}$,

$$\mathbb{P}_A(C) = \frac{\mathbb{P}(C \cap A)}{\mathbb{P}(A)} = \int_C \frac{\mathbf{1}_A}{\mathbb{P}(A)}\, d\mathbb{P}.$$

Thus

$$\frac{d\mathbb{P}_A}{d\mathbb{P}} = \frac{\mathbf{1}_A}{\mathbb{P}(A)} \quad \mathbb{P}\text{-almost surely.}$$

The two requested ledger rows are:

Representative	Numerator measure	Reference	Domain	Uniqueness
$d\mathbb{P}_A/d\mathbb{P}$	$C \mapsto \mathbb{P}(C \cap A)/\mathbb{P}(A)$	$\mathbb{P}$	$(\Omega, \mathcal{F})$	$\mathbb{P}$-a.s.
$\mathbb{E}[Y \mid \mathcal{G}]$	$C \mapsto \mathbb{E}[Y\mathbf{1}_C]$, $C \in \mathcal{G}$	$\mathbb{P}\vert_{\mathcal{G}}$	$(\Omega, \mathcal{G})$	$\mathbb{P}$-a.s.

If X has a continuous law, $\mathbb{P}(X = x) = 0$ for each fixed x. The event-ratio denominator therefore vanishes. A conditional kernel or density may provide a version, but it is a different construction and is not determined pointwise everywhere by the joint law.

Exercise 5 – Original deployment analogues

(a) Capped-payment mechanism. The transformed payment is $X \wedge 1$. The tail formula gives

$$\mathbb{E}[X \wedge 1] = \int_0^1 \mathbb{P}(X > t)\, dt = \int_0^1 (1+t)^{-2}\, dt = \left[-\frac{1}{1+t}\right]_0^1 = \frac{1}{2}.$$

(b) Nonlinear Poisson moment. For $N \sim \text{Poisson}(2)$,

$$\mathbb{E}[N] = 2, \qquad \text{Var}(N) = 2, \qquad \mathbb{E}[N^2] = 2 + 2^2 = 6.$$

Hence

$$\mathbb{E}\left[\frac{N(N+1)}{2}\right] = \frac{\mathbb{E}[N^2] + \mathbb{E}[N]}{2} = \frac{6+2}{2} = 4.$$

The required object is $\mathbb{E}[g(N)]$; $g(\mathbb{E}[N]) = 3$ is a different quantity.

(c) Finite-mixture inversion. The joint masses are

$$\mathbb{P}(A \cap B_1) = 0.3 \cdot 0.8 = 0.24, \qquad \mathbb{P}(A \cap B_2) = 0.7 \cdot 0.2 = 0.14.$$

Thus $\mathbb{P}(A) = 0.38$ and

$$\mathbb{P}(B_1 \mid A) = \frac{0.24}{0.38} = \frac{12}{19}.$$

Closed-book reconstruction

The first chain is

$$X \longmapsto \mu_X = X_{\#}\mathbb{P} \longmapsto \mathbb{E}[\varphi(X)] = \int_S \varphi\, d\mu_X.$$

No density is needed. A pdf appears only after choosing a reference λ and verifying $\mu_X \ll \lambda$.

For a finite partition, $\mathcal{G}$-measurability forces atomwise constants; the identities

$$\int_{B_i} H\, d\mathbb{P} = \int_{B_i} Y\, d\mathbb{P}$$

determine those constants on positive-probability atoms. Summation yields total expectation, and normalization of the corresponding joint masses yields Bayes' formula. Null atoms retain arbitrary coefficients because conditional expectation is defined only up to almost-sure equality.